

You are going to investigate the effect of the concentration of salt solutions on osmosis in potato tissue.

Read all the instructions but DO NOT DO THEM until you have drawn a table for your results in the space provided in 1(a)(ii).

You should use the safety equipment provided while you are doing the practical work.

You are going to use three different concentrations of salt solution.

Step 1 Label three beakers **S1**, **S2** and **S3**.

(a) (i) Decide the volumes of salt solution **S** and distilled water **W** you need to make 50 cm³ of a 0.5 mol per dm³ salt solution.

Complete Table 1.1 by writing in the volumes of **S** and **W** you will use to make the 0.5 mol per dm³ salt solution.

Table 1.1

beaker	volume of 1.0 mol per dm ³ salt solution S /cm ³	volume of distilled water W /cm ³	final concentration of salt solution /mol per dm ³
S1	50	0	1.0
S2			0.5
S3	0	50	0.0

[1]

Step 2 Use the volumes of 1.0 mol per dm³ salt solution **S** and distilled water **W** shown in Table 1.1 to make the salt solutions in beakers **S1**, **S2** and **S3**.

Step 3 Put the potato cylinders onto a white tile and cut each potato cylinder into 3 mm discs, as shown in Fig. 1.1.

You will need 30 discs of potato for this investigation and one extra potato disc for Question 1(b). Leave the extra potato disc on the white tile.



Fig. 1.1

Step 4 Place 10 potato discs in a line on the white tile. Place the ruler at the start of the line so you can measure the total length of the line of potato discs, as shown in Fig. 1.2.

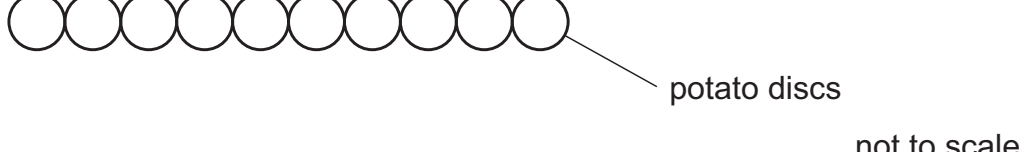


Fig. 1.2

Step 5 Measure the total length of the 10 potato discs and record this value in your table in 1(a)(ii).

Step 6 Put the 10 potato discs from step 5 into beaker **S1**.

Step 7 Repeat step 4 to step 6 with the other potato discs and the salt solutions in beakers **S2** and **S3**.

Step 8 Start the stop-clock and leave the potato discs in the beakers of salt solution for 20 minutes.

Continue with the rest of the questions while you are waiting.

Step 9 After 20 minutes, pour the salt solution from beaker **S1** into the beaker labelled **waste**.

Step 10 Put the 10 potato discs from beaker **S1** onto the white tile and line them up as shown in Fig. 1.2.

Step 11 Measure the final total length of the 10 potato discs and record this value in your table in 1(a)(ii).

Step 12 Pour the salt solution from beaker **S2** into the waste beaker and repeat step 10 and step 11.

Step 13 Pour the salt solution from beaker **S3** into the waste beaker and repeat step 10 and step 11.

(ii) Prepare a table and record your results. Include the final concentrations of the salt solutions in your table.

[4]

(iii) Calculate the **change** in total length of the 10 discs in each beaker.

beaker **S1**

beaker **S2**

beaker **S3**

[1]

(iv) State a conclusion for your results.

.....
.....
..... [1]

(v) State the dependent variable in this investigation.

..... [1]

(vi) Identify **one** source of error in this investigation.

.....

..... [1]

(vii) Describe **one** safety precaution you took when preparing the potato discs.

.....

..... [1]

(b) Put a few drops of iodine solution onto the potato disc that you kept from step 3.

Record the final colour of the iodine solution on the potato disc and state a conclusion for this test.

final colour

conclusion [1]

(c) Fig. 1.3 shows a leaf from a potato plant.

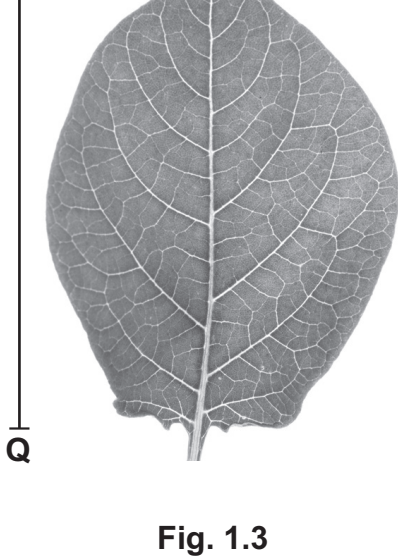


Fig. 1.3

(i) Draw a large diagram of the leaf shown in Fig. 1.3.

[4]

(ii) Line **PQ** represents the length of the leaf in Fig. 1.3.

Measure the length of line **PQ** in Fig. 1.3.

length of line **PQ** mm

The actual length of the leaf is 39 mm.

Using your measurement and the formula, calculate the magnification of the leaf in Fig. 1.3.

$$\text{magnification} = \frac{\text{length of line PQ in Fig. 1.3}}{\text{actual length of the leaf}}$$

Give your answer to **two** significant figures.

Space for working.

[3]

(iii) Fig. 1.4 shows a leaf from a tomato plant. The tomato leaf is narrower than the potato leaf shown in Fig. 1.3.



Fig. 1.4

Compare the leaves shown in Fig. 1.3 and in Fig. 1.4.

Describe **one** visible difference, other than size, and **one** visible similarity.

difference

.....

similarity

..... [2]

[Total: 20]